

Design and Analysis of a Routed Network Topology

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1. Introduction

This report presents the design and implementation of a routed network topology created using a network simulation environment. The topology demonstrates communication between multiple network segments using routers, switches, and end devices.

The network is designed to simulate a real- world enterprise environment where routing is used to allow communication between different IP networks. The topology highlights the importance of routers in directing traffic and ensuring connectivity between network segments.

2. Objectives

The main objectives of this topology are:

- To design a routed enterprise network.
- To understand router configuration and functionality.
- To enable communication between different networks.
- To configure IP addressing and gateways.
- To simulate inter- network communication.

3. Network Devices Used

Device Name	Device Type	Purpose
Router1	Router	Routes traffic between networks
Router2	Router	Connects additional network segments
Switch1	Layer- 2 Switch	Connects local devices within the LAN
Switch2	Layer- 2 Switch	Provides LAN connectivity
PC1	Personal Computer	Client workstation
PC2	Personal Computer	Client workstation
Server1	Server	Provides centralized network services

4. Network Topology Description

The topology consists of routers connected together to allow communication between separate network segments.

- Router1 connects to Switch1 and local devices.
- Router2 connects to Switch2 and additional hosts.
- The routers exchange routing information and forward packets between networks.
- PCs and servers communicate through the routing infrastructure.

This structure demonstrates how routing enables connectivity between multiple LANs.

5. Routing Functionality

Routers are responsible for forwarding packets between different IP networks.

Main router functions include:

- Determining the best path for data transmission.
- Connecting separate network segments.
- Managing routing tables.
- Reducing broadcast traffic.
- Improving network scalability.

Routing allows devices in different networks to communicate successfully.

6. Switch Functionality

The switches provide local area network connectivity by forwarding Ethernet frames based on MAC addresses.

Functions of switches include:

- Connecting end devices within the same LAN.
 - Improving network efficiency.
 - Supporting high-speed communication.
 - Reducing collisions within the network.
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7. IP Addressing and Communication

Each device within the topology is assigned an IP address according to its network segment. Routers use these addresses to forward packets between networks.

Communication process:

1. A device sends data to its default gateway.
 2. The router receives the packet.
 3. The router checks its routing table.
 4. The packet is forwarded to the destination network.
 5. The destination device receives the data.
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8. Advantages of the Topology

This routed network topology provides several advantages:

- Communication between multiple networks.
 - Better traffic management.
 - Improved scalability.
 - Enhanced organization of network segments.
 - Efficient packet forwarding.
 - Support for enterprise- level networking.
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9. Practical Applications

This type of topology is commonly used in:

- Enterprise networks.
 - Educational institutions.
 - Data centers.
 - Corporate branch networks.
 - Internet service provider infrastructures.
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10 . Conclusion

The implemented routed network topology successfully demonstrates how routers, switches, servers, and client devices interact within a multi- network environment.

The topology highlights the importance of routing in enabling communication between different network segments while improving scalability and network management. This practical implementation helps in understanding routing concepts, inter- network communication, and enterprise network design.